

SSILP Math

Beginner

VECTORS IN 2D
DOT PRODUCT



Session outline

1. Vectors in 2D



2. The Dot product

Learning objectives

- Understand the concept of vectors in 2D, including their geometric representation, magnitude, and direction.
- Learn how to perform vector addition and subtraction using geometric rules such as the parallelogram law.
- Represent vectors in coordinate form as ordered pairs or column vectors
- Understand dot product, and its applications in physics and engineering.
 - Analyze the perpendicularity of vectors using the dot product and understand the concept of scalar projection.
 - Calculate work in physics using vector operations, including dot products of force and displacement vectors.

1. Vectors in 2D



Vectors in two and three dimensions

Overview

Vectors in two
dimensions

The dot product

Vectors in 3D

Cross product

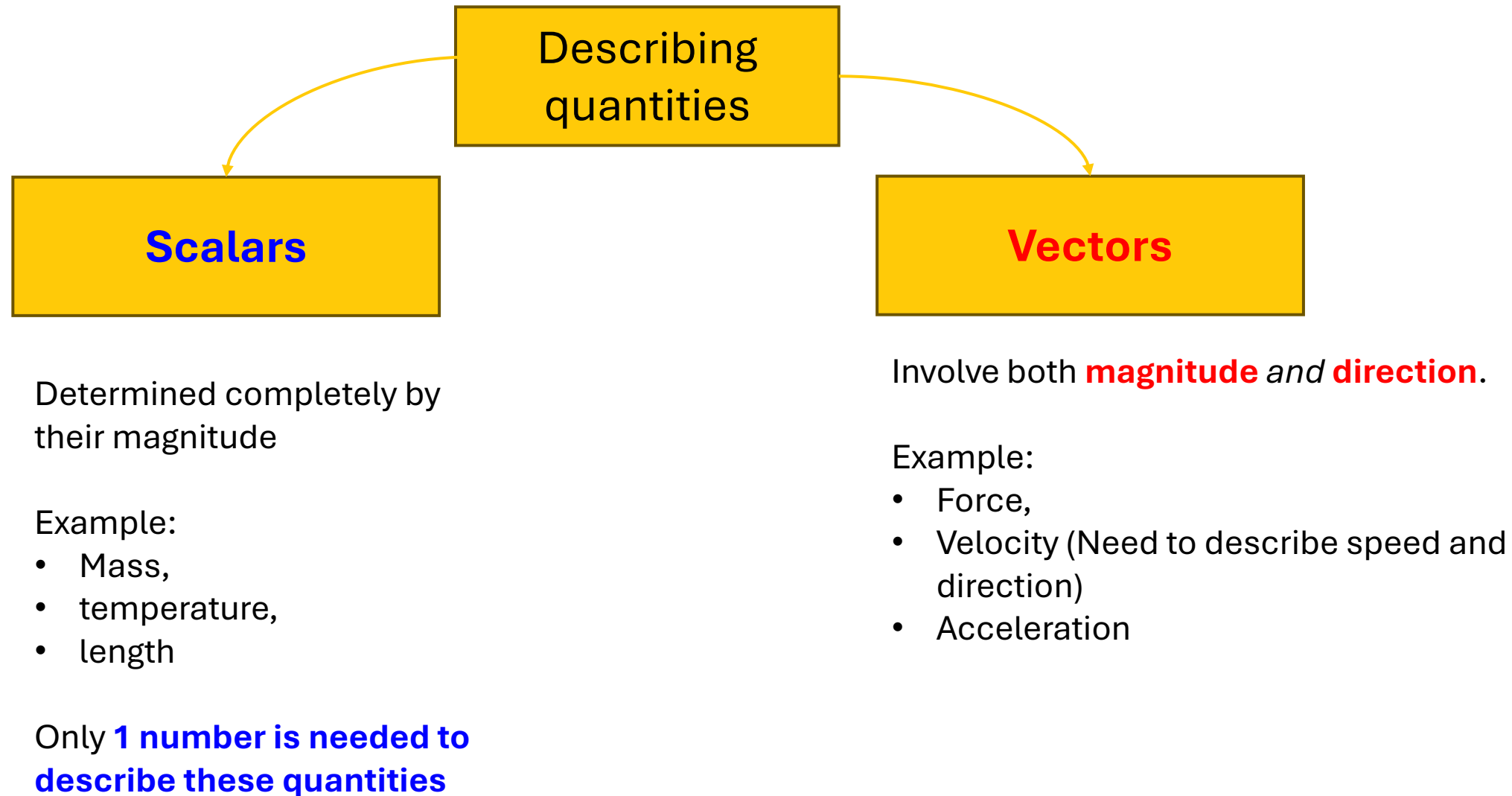
Equations of lines
and planes

Textbook Reference

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 9,
Sections 9.1 to 9.6

What are vectors?



Why are vectors important?

7

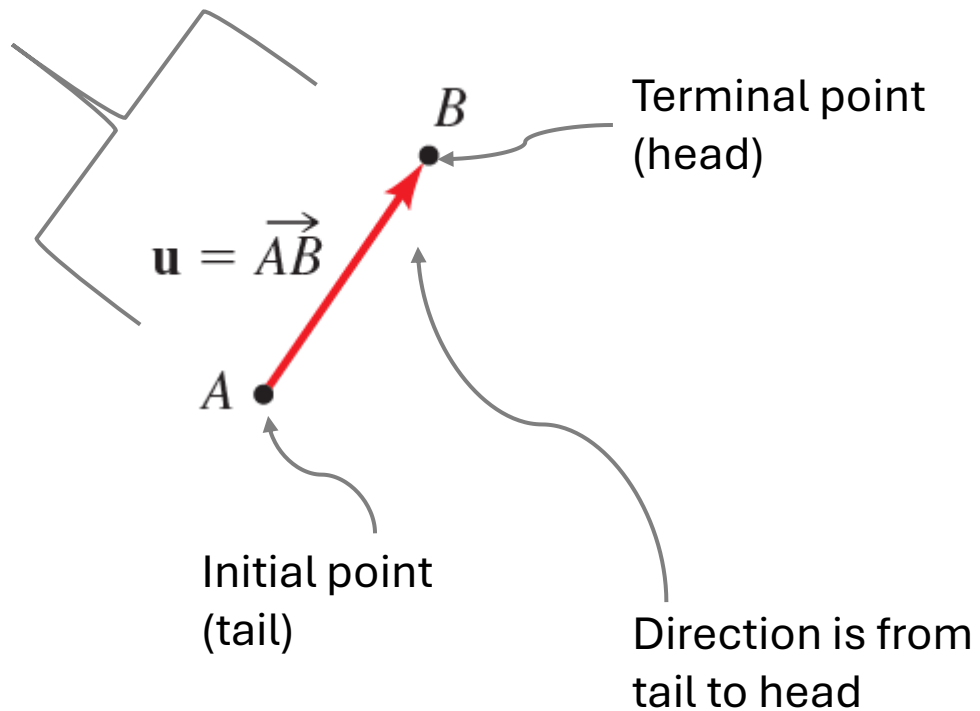
Vectors are the building blocks of multivariable calculus

Used to represent coordinates in higher dimensional space

Used in computer science to represents lists

Geometric description of vectors

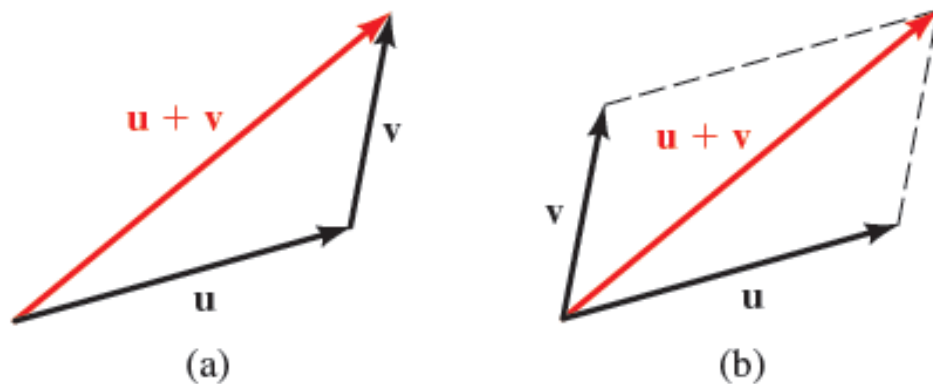
Length or
magnitude



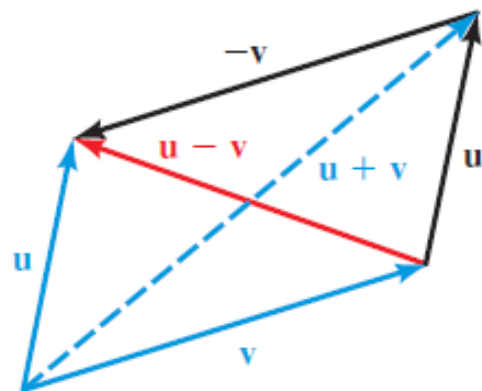
The **length of the line segment AB**
is called the **magnitude** or **length** of the vector
and is denoted by

$$|\overrightarrow{AB}|$$

Vector addition and subtraction



If we draw $\mathbf{u}$ and $\mathbf{v}$ starting at the same point, then $\mathbf{u} + \mathbf{v}$ is the vector that is the diagonal of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$



The **difference** of two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined by $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

Vector notation

Boldface (as in the textbook): $\mathbf{u}, \mathbf{v}$

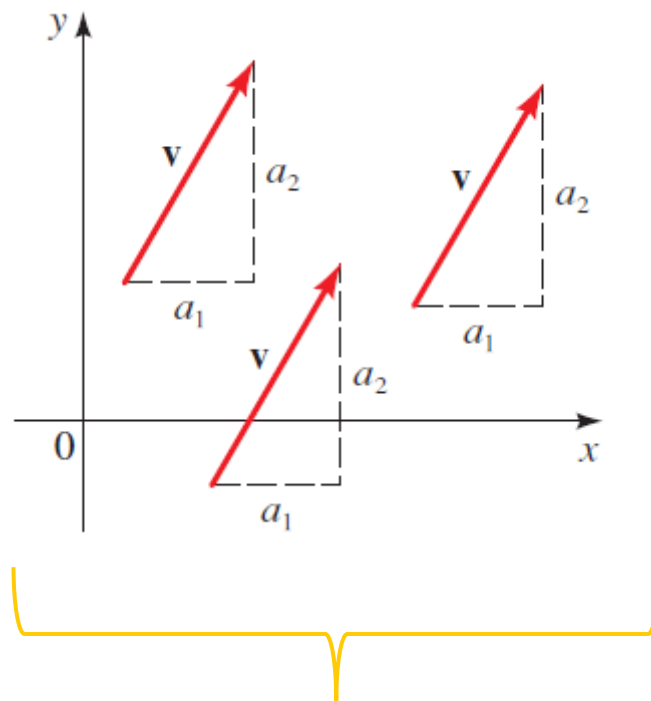
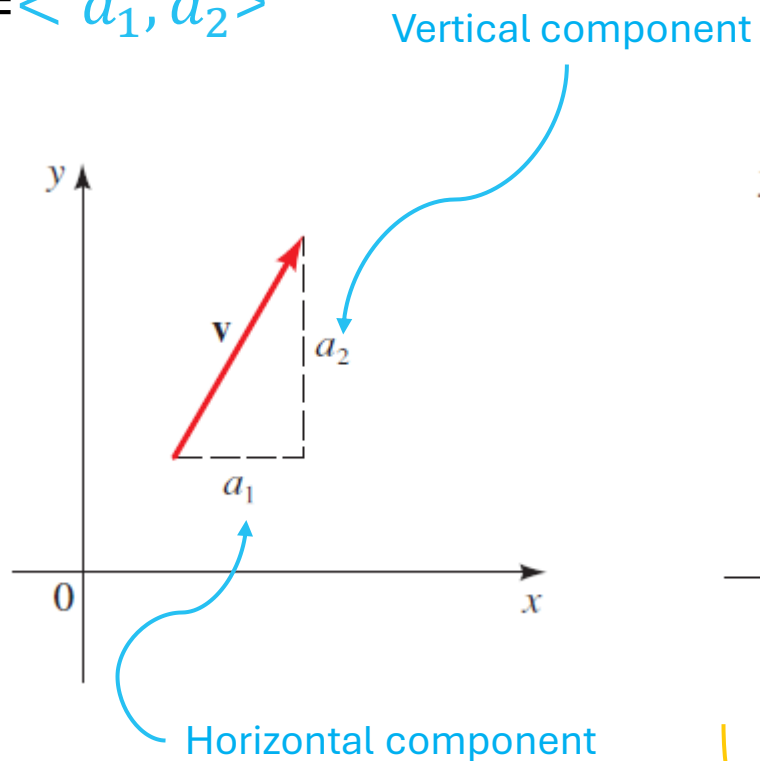
Or

With an arrow on top: $\vec{u}, \vec{v}$

Vectors in the coordinate plane

We represent vector $\mathbf{v}$ as an **ordered pair of real numbers**

$$\mathbf{v} = \langle a_1, a_2 \rangle$$



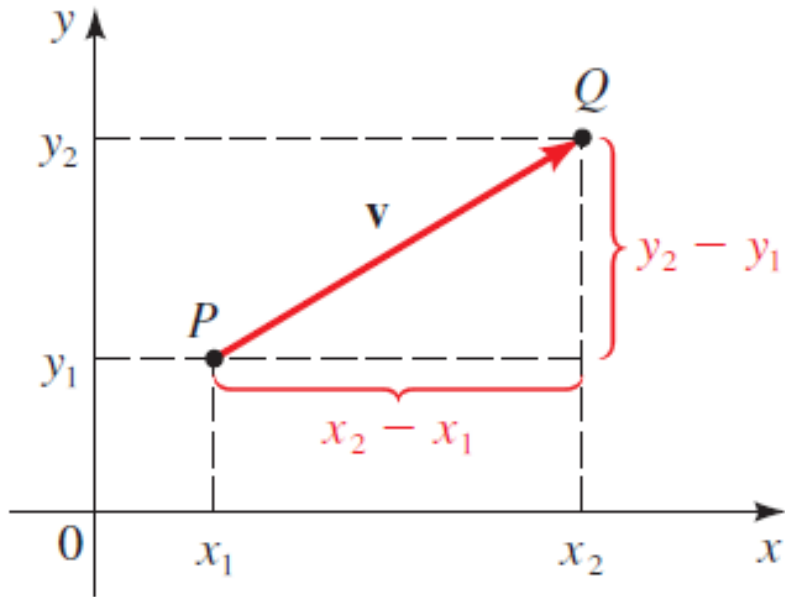
vector $\langle a_1, a_2 \rangle$ has
many different representations,
 depending on its initial point

Component form of a vector

COMPONENT FORM OF A VECTOR

If a vector $\mathbf{v}$ is represented in the plane with initial point $P(x_1, y_1)$ and terminal point $Q(x_2, y_2)$, then

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$$



Alternate notations for component form

The following notations are also used to represent the component form

$$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ (column vector notation)}$$

$$\vec{v} = [x \ y] \text{ (row vector notation)}$$

$$\vec{v} = x\hat{i} + y\hat{j} \text{ (unit vector notation)}$$

where

$x = x_2 - x_1$ is the horizontal component of $\vec{v}$ and
 $y = y_2 - y_1$ is the vertical component of $\vec{v}$.

Describing Vectors in Component Form

- (a) Find the component form of the vector $\mathbf{u}$ with initial point $(-2, 5)$ and terminal point $(3, 7)$.
- (b) If the vector $\mathbf{v} = \langle 3, 7 \rangle$ is sketched with initial point $(2, 4)$, what is its terminal point?
- (c) Sketch representations of the vector $\mathbf{w} = \langle 2, 3 \rangle$ with initial points at $(0, 0)$, $(2, 2)$, $(-2, -1)$, and $(1, 4)$.

SOLUTION

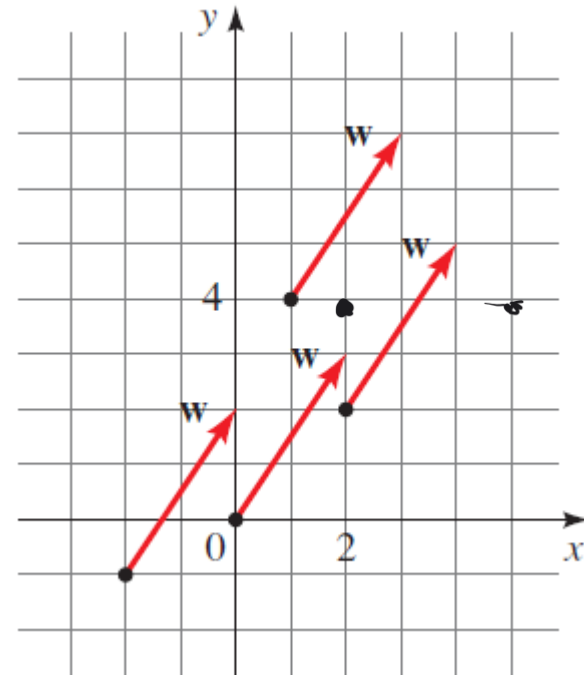
- (a) The desired vector is

$$\mathbf{u} = \langle 3 - (-2), 7 - 5 \rangle = \langle 5, 2 \rangle$$

- (b) Let the terminal point of $\mathbf{v}$ be (x, y) . Then

$$\langle x - 2, y - 4 \rangle = \langle 3, 7 \rangle$$

So $x - 2 = 3$ and $y - 4 = 7$, or $x = 5$ and $y = 11$. The terminal point is $(5, 11)$.



EXAMPLE

Describing Vectors in component form

13

(a) $P(1, 1), Q(9, 9)$

Here P is the initial point and Q is the terminal point

(b) $P(-1, 3), Q(-6, -1)$

ACTIVITY 1.1

1.1 Describing Vectors in component form

Sketching vectors

(a) Sketch the given vector with initial point $(4, 3)$, and find the terminal point

$$\mathbf{u} = \langle -8, -1 \rangle$$

(b) Sketch representations of the given vector with initial points at $(0, 0)$, $(2, 3)$ and $(-3, 5)$

$$\mathbf{u} = \langle 4, -6 \rangle$$

1.2a Describing Vectors in component form

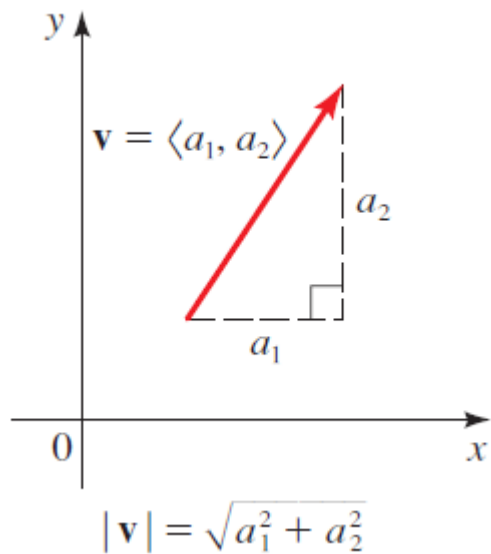
1.2b Describing Vectors in component form

Magnitude of a vector

MAGNITUDE OF A VECTOR

The **magnitude** or **length** of a vector $\mathbf{v} = \langle a_1, a_2 \rangle$ is

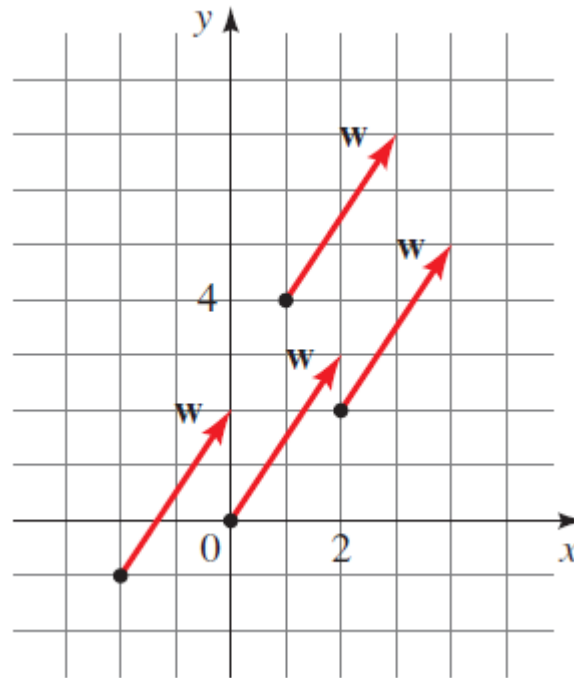
$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2}$$



Confusion alert!

The notation stands for
magnitude here,
not absolute value sign!

Two vectors are
equal
if and only if
their
corresponding
components are
equal



Magnitude of vector

Find the magnitude of each vector.

(a) $\mathbf{u} = \langle 2, -3 \rangle$ (b) $\mathbf{v} = \langle 5, 0 \rangle$ (c) $\mathbf{w} = \langle \frac{3}{5}, \frac{4}{5} \rangle$

SOLUTION

(a) $|\mathbf{u}| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

(b) $|\mathbf{v}| = \sqrt{5^2 + 0^2} = \sqrt{25} = 5$

(c) $|\mathbf{w}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1$

Algebraic operations on vectors

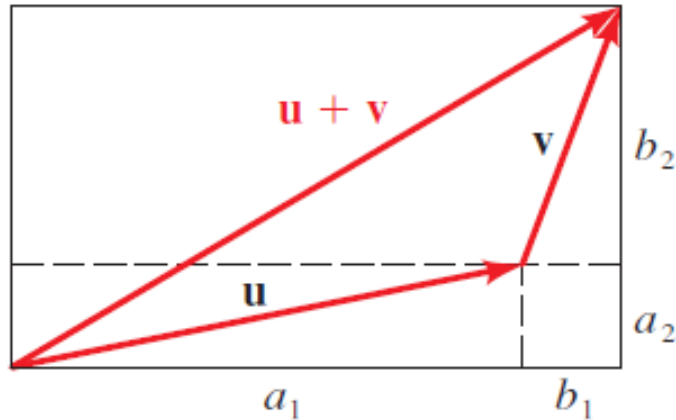
ALGEBRAIC OPERATIONS ON VECTORS

If $\mathbf{u} = \langle a_1, a_2 \rangle$ and $\mathbf{v} = \langle b_1, b_2 \rangle$, then

$$\mathbf{u} + \mathbf{v} = \langle a_1 + b_1, a_2 + b_2 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle a_1 - b_1, a_2 - b_2 \rangle$$

$$c\mathbf{u} = \langle ca_1, ca_2 \rangle \quad c \in \mathbb{R}$$



Column vector notation

$$\mathbf{u} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$\mathbf{v} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix}$$

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} a_1 - b_1 \\ a_2 - b_2 \end{bmatrix}$$

$$c\mathbf{u} = \begin{bmatrix} ca_1 \\ ca_2 \end{bmatrix}$$

Properties of vectors

PROPERTIES OF VECTORS

Vector addition

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$$

Length of a vector

$$|c\mathbf{u}| = |c| |\mathbf{u}|$$

Multiplication by a scalar

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$(cd)\mathbf{u} = c(d\mathbf{u}) = d(c\mathbf{u})$$

$$1\mathbf{u} = \mathbf{u}$$

$$0\mathbf{u} = \mathbf{0}$$

$$c\mathbf{0} = \mathbf{0}$$

Note: this is a **zero vector**, or vector in space that has a magnitude equal to 0 and an undefined direction.

Zero vector is given by $\vec{0} = \langle 0, 0, 0 \rangle$ in 3D space and $\vec{0} = \langle 0, 0 \rangle$ in 2D space

Operations on vectors

If $\mathbf{u} = \langle 2, -3 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$, find $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $2\mathbf{u}$, $-3\mathbf{v}$, and $2\mathbf{u} + 3\mathbf{v}$.

SOLUTION By the definitions of the vector operations we have

$$\mathbf{u} + \mathbf{v} = \langle 2, -3 \rangle + \langle -1, 2 \rangle = \langle 1, -1 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle 2, -3 \rangle - \langle -1, 2 \rangle = \langle 3, -5 \rangle$$

$$2\mathbf{u} = 2\langle 2, -3 \rangle = \langle 4, -6 \rangle$$

$$-3\mathbf{v} = -3\langle -1, 2 \rangle = \langle 3, -6 \rangle$$

$$2\mathbf{u} + 3\mathbf{v} = 2\langle 2, -3 \rangle + 3\langle -1, 2 \rangle = \langle 4, -6 \rangle + \langle -3, 6 \rangle = \langle 1, 0 \rangle$$

Unit vector

A vector of length 1 is called a **unit vector**.

Two useful unit vectors are $\mathbf{i}$ and $\mathbf{j}$, defined by

$$\mathbf{i} = \langle 1, 0 \rangle \quad \mathbf{j} = \langle 0, 1 \rangle$$

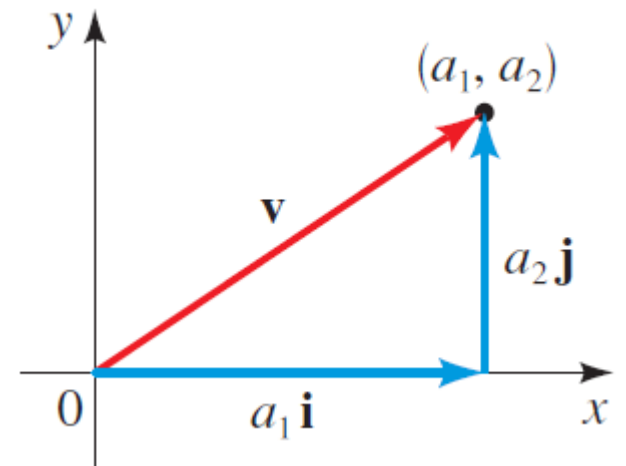
What's so special about unit vectors?

Any vector can be expressed in terms of unit vectors!

VECTORS IN TERMS OF $\mathbf{i}$ AND $\mathbf{j}$

The vector $\mathbf{v} = \langle a_1, a_2 \rangle$ can be expressed in terms of $\mathbf{i}$ and $\mathbf{j}$ by

$$\mathbf{v} = \langle a_1, a_2 \rangle = a_1 \mathbf{i} + a_2 \mathbf{j}$$



Vectors in terms of $\mathbf{i}$ and $\mathbf{j}$

(a) Write the vector $\mathbf{u} = \langle 5, -8 \rangle$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

(b) If $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = -\mathbf{i} + 6\mathbf{j}$, write $2\mathbf{u} + 5\mathbf{v}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

SOLUTION

(a) $\mathbf{u} = 5\mathbf{i} + (-8)\mathbf{j} = 5\mathbf{i} - 8\mathbf{j}$

(b) The properties of addition and scalar multiplication of vectors show that we can manipulate vectors in the same way as algebraic expressions. Thus

$$\begin{aligned} 2\mathbf{u} + 5\mathbf{v} &= 2(3\mathbf{i} + 2\mathbf{j}) + 5(-\mathbf{i} + 6\mathbf{j}) \\ &= (6\mathbf{i} + 4\mathbf{j}) + (-5\mathbf{i} + 30\mathbf{j}) \\ &= \mathbf{i} + 34\mathbf{j} \end{aligned}$$

Horizontal and vertical components of a vector

HORIZONTAL AND VERTICAL COMPONENTS OF A VECTOR

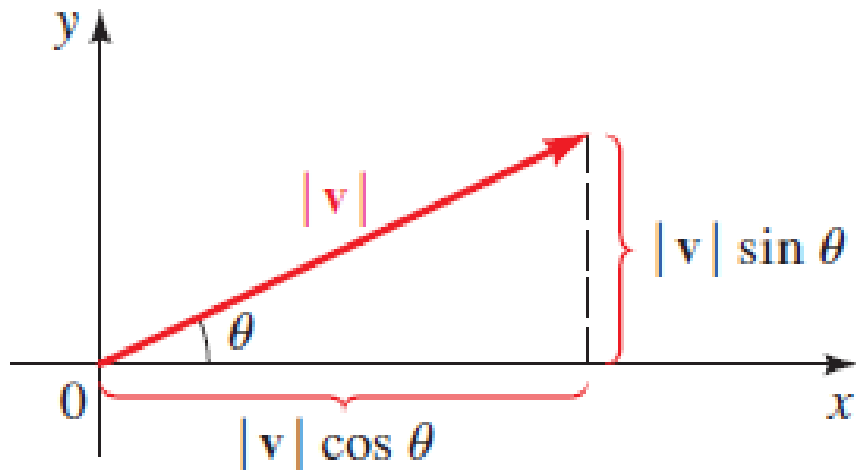
Let $\mathbf{v}$ be a vector with magnitude $|\mathbf{v}|$ and direction θ .

Then $\mathbf{v} = \langle a_1, a_2 \rangle = a_1 \mathbf{i} + a_2 \mathbf{j}$, where

$$a_1 = |\mathbf{v}| \cos \theta \quad \text{and} \quad a_2 = |\mathbf{v}| \sin \theta$$

Thus we can express $\mathbf{v}$ as

$$\mathbf{v} = |\mathbf{v}| \cos \theta \mathbf{i} + |\mathbf{v}| \sin \theta \mathbf{j}$$



Magnitude of vector

Find $|\mathbf{u}|$, $|\mathbf{v}|$, $|2\mathbf{u}|$, $|\frac{1}{2}\mathbf{v}|$, $|\mathbf{u} + \mathbf{v}|$, $|\mathbf{u} - \mathbf{v}|$, and $|\mathbf{u}| - |\mathbf{v}|$

$$\mathbf{u} = -2\mathbf{i} + 3\mathbf{j}, \quad \mathbf{v} = \mathbf{i} - 2\mathbf{j}$$

ACTIVITY 1.3

Magnitude of vector

2. The Dot product

Overview

**The Dot Product of
Vectors**

**The Component of u
along v**

Work

Dot product: Intuitive understanding

The dot product, also called scalar product, is **a measure of how closely two vectors align**, in terms of the directions they point.

The measure is a **scalar number (single value)** that can be used to **compare the two vectors** and **to understand** the impact of repositioning one or both of them.

application

Example: Designing a video game

- *Are two characters looking in the same direction?*
 - *Is a character looking at a sign?*
- *How should a ray of light interact with a surface?*

We answer these types of questions with the **dot product**.

Dot product: Formal definition

DEFINITION OF THE DOT PRODUCT

If $\mathbf{u} = \langle a_1, a_2 \rangle$ and $\mathbf{v} = \langle b_1, b_2 \rangle$ are vectors, then their **dot product**, denoted by $\mathbf{u} \cdot \mathbf{v}$, is defined by

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2$$

We are just multiplying corresponding components of the two vectors and adding!

“The Dot Product” is especially useful in calculus and in applications of vectors to physics and engineering.

Important

The dot product is ***not a vector***; it is a real number, or **scalar**

Calculating Dot product

(a) If $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle 4, 5 \rangle$ then

$$\mathbf{u} \cdot \mathbf{v} = (3)(4) + (-2)(5) = 2$$

(b) If $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = 5\mathbf{i} - 6\mathbf{j}$, then

$$\mathbf{u} \cdot \mathbf{v} = (2)(5) + (1)(-6) = 4$$

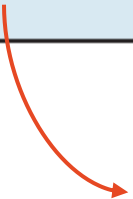
EXAMPLE

Properties of Dot product

PROPERTIES OF THE DOT PRODUCT

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
3. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
4. $|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u}$

How did we
get this?


$$\mathbf{u} \cdot \mathbf{u} = a_1a_1 + a_2a_2 = a_1^2 + a_2^2 = |\mathbf{u}|^2$$

Dot product theorem

THE DOT PRODUCT THEOREM

If θ is the angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

ANGLE BETWEEN TWO VECTORS

If θ is the angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, then

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

Dot products are a way of combining two vectors.

“How much the vectors are pointing in the same direction”

→ We consider the magnitude of the two vectors and the angle between them.

Finding the Angle Between Two Vectors

Find the angle between the vectors $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle 4, -3 \rangle$.

SOLUTION By the formula for the angle between two vectors we have

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} = \frac{(2)(4) + (5)(-3)}{\sqrt{4 + 25} \sqrt{16 + 9}} = \frac{-7}{5\sqrt{29}}$$

Thus the angle between $\mathbf{u}$ and $\mathbf{v}$ is

$$\theta = \cos^{-1}\left(\frac{-7}{5\sqrt{29}}\right) \approx 105.1^\circ$$

EXAMPLE

Calculating dot product

Find $\mathbf{u} \cdot \mathbf{v}$ and the angle between the vectors

(a) $\mathbf{u} = \mathbf{i} + \sqrt{3}\mathbf{j}, \quad \mathbf{v} = -\sqrt{3}\mathbf{i} + \mathbf{j}$

(b) $\mathbf{u} = \langle -6, 6 \rangle, \quad \mathbf{v} = \langle 1, -1 \rangle$

(c) $\mathbf{u} = -5\mathbf{j}, \quad \mathbf{v} = -\mathbf{i} - \sqrt{3}\mathbf{j}$

ACTIVITY 2.1

2.1(a) Calculating dot product

2.1(b) Calculating dot product

SOLUTION

2.1(c) Calculating dot product

SOLUTION

Orthogonal vectors

ORTHOGONAL VECTORS

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

If $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, then the angle between them is $\pi/2$, so

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \frac{\pi}{2} = 0$$

Conversely, if $\mathbf{u} \cdot \mathbf{v} = 0$, then

$$|\mathbf{u}| |\mathbf{v}| \cos \theta = 0$$

Since $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors, we conclude that $\cos \theta = 0$, so $\theta = \pi/2$. Thus $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

Since

$$\mathbf{0} \cdot \vec{x} = \vec{0}$$

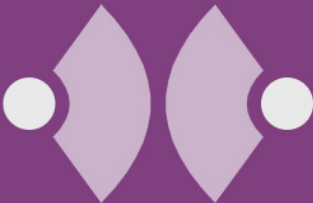
for any vector x ,

**zero vector is orthogonal
to every vector**

Vector directions and dot product: Think Intuitively

The dot product answers the question:

how similar are the directions of two vectors?

	Same Directions = 1
	Opposite Directions = -1
	Perpendicular Directions = 0

Dot product is largest as
 $\cos 0 = 1$

Dot product is negative magnitudes of two vectors
 $\cos 180 = -1$

Dot product is zero as
 $\cos \frac{\pi}{2} = 0$

Checking whether two vectors are perpendicular

Determine whether the vectors in each pair are perpendicular.

(a) $\mathbf{u} = \langle 3, 5 \rangle$ and $\mathbf{v} = \langle 2, -8 \rangle$ (b) $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$

SOLUTION

(a) $\mathbf{u} \cdot \mathbf{v} = (3)(2) + (5)(-8) = -34 \neq 0$, so $\mathbf{u}$ and $\mathbf{v}$ are not perpendicular.

(b) $\mathbf{u} \cdot \mathbf{v} = (2)(-1) + (1)(2) = 0$, so $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

EXAMPLE

Perpendicular vectors?

Determine whether the given vectors are perpendicular.

(a) $\mathbf{u} = \langle 0, -5 \rangle, \quad \mathbf{v} = \langle 4, 0 \rangle$

(b) $\mathbf{u} = 2\mathbf{i}, \quad \mathbf{v} = -7\mathbf{j}$

ACTIVITY 2.2

2.2 Perpendicular vectors?

SOLUTION

The component of $\mathbf{u}$ along $\mathbf{v}$

The **component of $\mathbf{u}$ along $\mathbf{v}$** (also called the **component of $\mathbf{u}$ in the direction of $\mathbf{v}$** or the **scalar projection of $\mathbf{u}$ onto $\mathbf{v}$**) is defined to be

$$|\mathbf{u}| \cos \theta$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

Intuitively, the component of $\mathbf{u}$ along $\mathbf{v}$ is the magnitude of the portion of $\mathbf{u}$ that points in the direction of $\mathbf{v}$

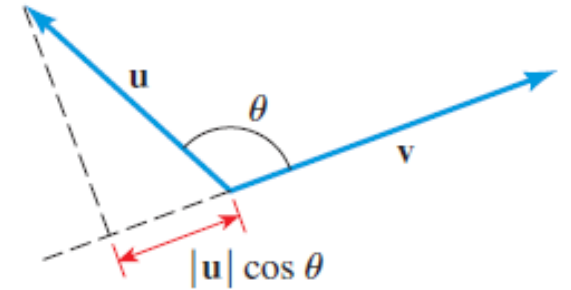
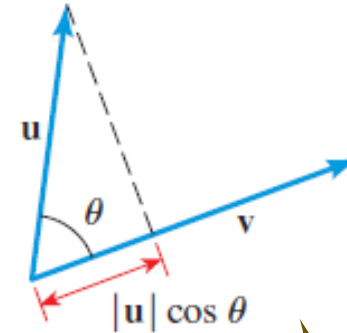
The component of $\mathbf{u}$ along $\mathbf{v}$ can be computed by using dot products:

$$|\mathbf{u}| \cos \theta = \frac{|\mathbf{v}| |\mathbf{u}| \cos \theta}{|\mathbf{v}|} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$$

THE COMPONENT OF $\mathbf{u}$ ALONG $\mathbf{v}$

The component of $\mathbf{u}$ along $\mathbf{v}$ (or the scalar projection of $\mathbf{u}$ onto $\mathbf{v}$) is

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$$



the component of $\mathbf{u}$
along $\mathbf{v}$ is negative
if

$$\frac{\pi}{2} < \theta \leq \pi$$

How long is
the shadow of
 $\mathbf{u}$ along $\mathbf{v}$

The component of $\mathbf{u}$ along $\mathbf{v}$

Find the component of $\mathbf{u}$ along $\mathbf{v}$

(a) $\mathbf{u} = \langle -3, 5 \rangle, \quad \mathbf{v} = \langle 1/\sqrt{2}, 1/\sqrt{2} \rangle$

(b) $\mathbf{u} = 7\mathbf{i}, \quad \mathbf{v} = 8\mathbf{i} + 6\mathbf{j}$

ACTIVITY 2.3

2.3a The component of u along v

SOLUTION

2.3b The component of u along v

SOLUTION

In everyday use, the term **work** means the total amount of effort required to perform a task.

In physics, *work* has a technical meaning that conforms to this intuitive meaning.

If a constant **force of magnitude F** moves an object through a **distance d** along a straight line, then the **work** done is

$$\text{Work} = \text{force} \times \text{distance}$$

$$W = Fd$$



This formula
applies
only when
the force is directed
along the direction
of motion.

If **F is measured in newtons** and **d in meters**, then the unit of work is a **newton-meter** (N-m).

WORK

The work W done by a force $\mathbf{F}$ in moving along a vector $\mathbf{D}$ is

$$W = \mathbf{F} \cdot \mathbf{D}$$

A force is given by the vector $\mathbf{F} = \langle 2, 3 \rangle$ and moves an object from the point $(1, 3)$ to the point $(5, 9)$. Find the work done.

SOLUTION The displacement vector is

$$\mathbf{D} = \langle 5 - 1, 9 - 3 \rangle = \langle 4, 6 \rangle$$

So the work done is

$$W = \mathbf{F} \cdot \mathbf{D} = \langle 2, 3 \rangle \cdot \langle 4, 6 \rangle = 26$$

If the unit of force is pounds and the distance is measured in feet, then the work done is 26 ft-lb.

Calculating work

A man pulls a wagon horizontally by exerting a force of 20 lb on the handle. If the handle makes an angle of 60° with the horizontal, find the work done in moving the wagon 100 ft.

SOLUTION We choose a coordinate system with the origin at the initial position of the wagon (see Figure 8). That is, the wagon moves from the point $P(0, 0)$ to the point $Q(100, 0)$. The vector that represents this displacement is

$$\mathbf{D} = 100\mathbf{i}$$

The force on the handle can be written in terms of components (see Section 9.1) as

$$\mathbf{F} = (20 \cos 60^\circ)\mathbf{i} + (20 \sin 60^\circ)\mathbf{j} = 10\mathbf{i} + 10\sqrt{3}\mathbf{j}$$

Thus the work done is

$$W = \mathbf{F} \cdot \mathbf{D} = (10\mathbf{i} + 10\sqrt{3}\mathbf{j}) \cdot (100\mathbf{i}) = 1000 \text{ ft-lb}$$

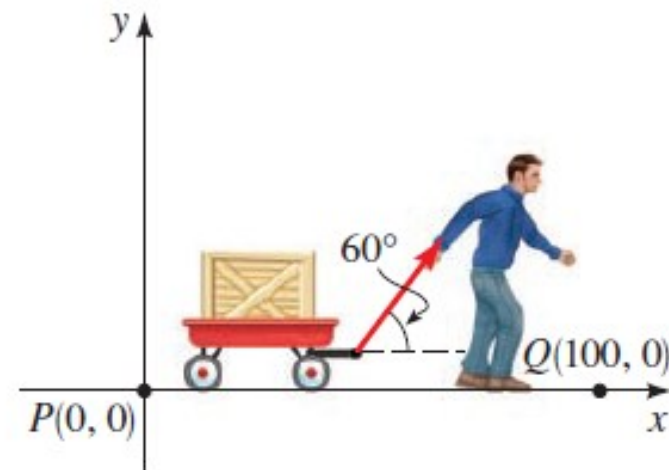


FIGURE 8

Calculating work

Find the work done by the force $\mathbf{F}$ in moving an object from P to Q .

(a) $\mathbf{F} = 400\mathbf{i} + 50\mathbf{j}; \quad P(-1, 1), Q(200, 1)$

(b) $\mathbf{F} = -4\mathbf{i} + 20\mathbf{j}; \quad P(0, 10), Q(5, 25)$

ACTIVITY 2.4

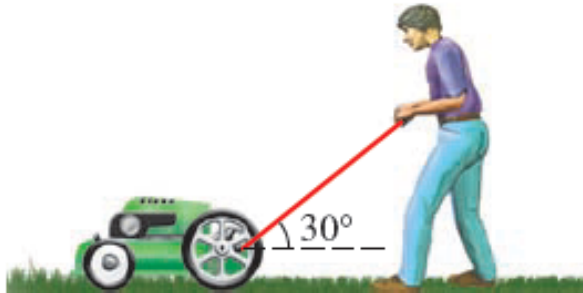
2.4 Calculating work

Word problems

- (1) **Work** A lawn mower is pushed a distance of 200 ft along a horizontal path by a constant force of 50 lb. The handle of the lawn mower is held at an angle of 30° from the horizontal (see the figure). Find the work done.

Use
60 m

Use
220 N



ACTIVITY

